

Loinsgate Estate Agency Portal Report

Project Purpose:

The purpose of my project was to design and develop a database-driven Real Estate Agency Portal using PHP and MySQL. The system allows users to register accounts, log in securely, browse properties, and manage real estate listings through a web-based platform.

The application supports the following user roles:

- Agent
- Buyer
- Renter

Agents are responsible for adding and managing property listings, a buyers and renters can browse listings and submit inquiries.

The project demonstrates:

- PHP programming
- MySQL database
- CRUD operations
- Authentication and sessions
- Role-based access control
- Relational database design
- Website customization and branding

Database Design:

The database created for my project is:

</> SQL

real_estate_portal_db

The database contains five main tables and one database view.

Users Table:

The Users table stores account information for all system users.

<u>Column</u>	<u>Description</u>
userId	Primary key
userName	Username
contactInfo	Email or phone
passwordHash	User password
userType	Agent, Buyer, or Renter

Properties Table:

The Properties table stores all property listings added by agents.

<u>Column</u>	<u>Description</u>
propertyId	Primary key
title	Property title
propertyType	Property category
address	Property address
city	Property city
price	Property price
status	Available, sold, rented
agentId	Foreign key linked to Users

Inquiries Table:

The Inquiries table stores messages submitted by users regarding properties.

<u>Column</u>	<u>Description</u>
inquiryId	Primary key
userId	User submitting inquiry
propertyId	Related property
message	Inquiry message
inquiryDate	Date submitted

Transactions Table:

The Transactions table stores rental and sale transaction information.

Column	Description
transactionId	Primary key
propertyId	Related property
userId	User involved
transactionType	Sale or rental
transactionDate	Transaction Date
amount	Transaction amount

Favorites Table:

The Favorites table stores properties saved by users.

Column	Description
favoriteId	Primary key
userId	User saving property
propertyId	Saved property
savedDate	Date saved

Database View:

The project also includes a SQL view named:

```
</> SQL
```

PropertyListingView

The view joins the Properties and Users tables to display property listings together with the agent's username.

Relationships:

Foreign key relationships were implemented to maintain database integrity:

- Users → Properties
- Users → Inquires
- Properties → Inquiries
- Properties → Transactions
- Users → Transactions
- Users → Favorites
- Properties → Favorites

PHP Functionality

My project was built using PHP and organized into multiple pages and reusable classes.

Main PHP Pages

File	Purpose
index.php	Homepage
register.php	User registration
login.php	User login
logout.php	Session logout
dashboard.php	User dashboard
properties.php	Browse properties
property_details.php	View property details
add_property.php	Add new property
delete_property.php	Delete property
submit_inquiry.php	Submit property inquiry

Classes and Includes

File	Purpose
Database.php	Database connection
RealEstateDatabase.php	CRUD database methods
auth.php	Session authentication
header.php	Shared website header
footer.php	Shared website footer

CRUD Operations

The system supports completed CRUD functionality.

Create

- Register users
- Add property listings
- Submit inquiries

Read

- Browse all properties
- View detailed listings

Update

- Property status updated through transactions.

Delete

- Agents can delete properties from the database

Login and Session Handling:

My project used PHP session management for authentication and security.

When a user logs in:

1. Username and password are verified
2. User information is retrieved from the database
3. Session variables are created

The sessions stores:

```
</> PHP
```

```
$_SESSION['user']
```

Stored sessions data includes:

- userID
- username
- userType

Session Protection

Protected pages use the authentication function:

```
</> PHP
```

```
requireLogin();
```

This prevents unauthorized access to:

- Dashboard
- Add Property
- Delete Property
- Other restricted pages

The logout system ends the session and redirects the user back to the login page.

Role-Bases Access Control

The project implements role-based access control to limit system permissions.

Agent Role

Agents can:

- Add properties
- Delete properties

- Manage listings
- Browse listings

Buyer Role

Buyers can:

- Browse properties
- Submit properties

Renter Role

Renters can:

- Browse properties
- Submit inquiries

Role Verification Example

The dashboard checks the logged-in user type before displaying agent-only features:

```
</> PHP
if ($_SESSION['user']['userType'] === 'agent')
```

This ensures that only agents can access management functions.

Delete Property Feature

One of the advanced features added to the project was the ability for agents to delete property listings.

A “Delete Property” link was added to the property listings page.

Delete Process

The system deletes:

1. Related favorites
2. Related inquiries
3. Related transactions
4. The property itself

This prevents foreign key constraint errors.

Delete Method

The delete functionality was implemented inside:

```
</> PHP
RealEstateDatabase.php
```

Using a transaction-based method:

```
</> PHP
deleteProperty()
```

The function uses:

- SQL prepared statements
- Transactions
- Rollback protection

This ensures database consistency if an error occurs during deletion.

Website Customization and Branding

The website was customized to create a more professional luxury real estate appearance.

Logo Customization

A custom logo was added for:

Loinsgate Real Estate

The logo was uploaded into the project and displayed in the website header.

Color Theme

The website was redesigned using:

- Dark navy-blue background
- Gold accent colors
- Modern luxury styling

The styling was added through:

```
</> CSS  
assets/style.css
```

Layout Improvements

Additional styling improvements included:

- Centered logo
- Customized navigation bar
- Property cards
- Improved typography
- Styled buttons and links
- Luxury real estate branding appearance

These changes transformed the original basic starter template into a life like commercial real estate website, which is a personal goal of mine to invest in real estate.

Challenges and Solutions

Challenge 1: Database Connection Errors

The website initially failed to connect to MySQL because of incorrect credentials and port settings.

Solution

The database configuration was updated to:

- localhost
- port 3307
- correct database name
- proper root password configuration

Challenge 2: Undefined Methods

Errors occurred because methods such as:

```
</> PHP  
  
addUser()  
getUserByUsername()  
deleteProperty()
```

were missing from the database class.

Solution

The missing methods were added using prepared SQL statements inside:

```
</> PHP  
RealEstateDatabase.php
```

Challenge 3: Session Authentication Errors

The dashboard originally failed because the authentication function was missing.

Solution

An auth.php file was created containing;

```
</> PHP  
requireLogin();
```

This secured all protected pages.

Challenge 4: Delete Property Foreign Key Errors

Property deletion originally failed because related records still existed in:

- Favorites
- Inquires
- Transactions

Solution

The delete function was updated to remove related records first before deleting the property. Transaction and rollback protection were added for database safety.

Challenge 5: Website Design

The started website was very plain with no color.

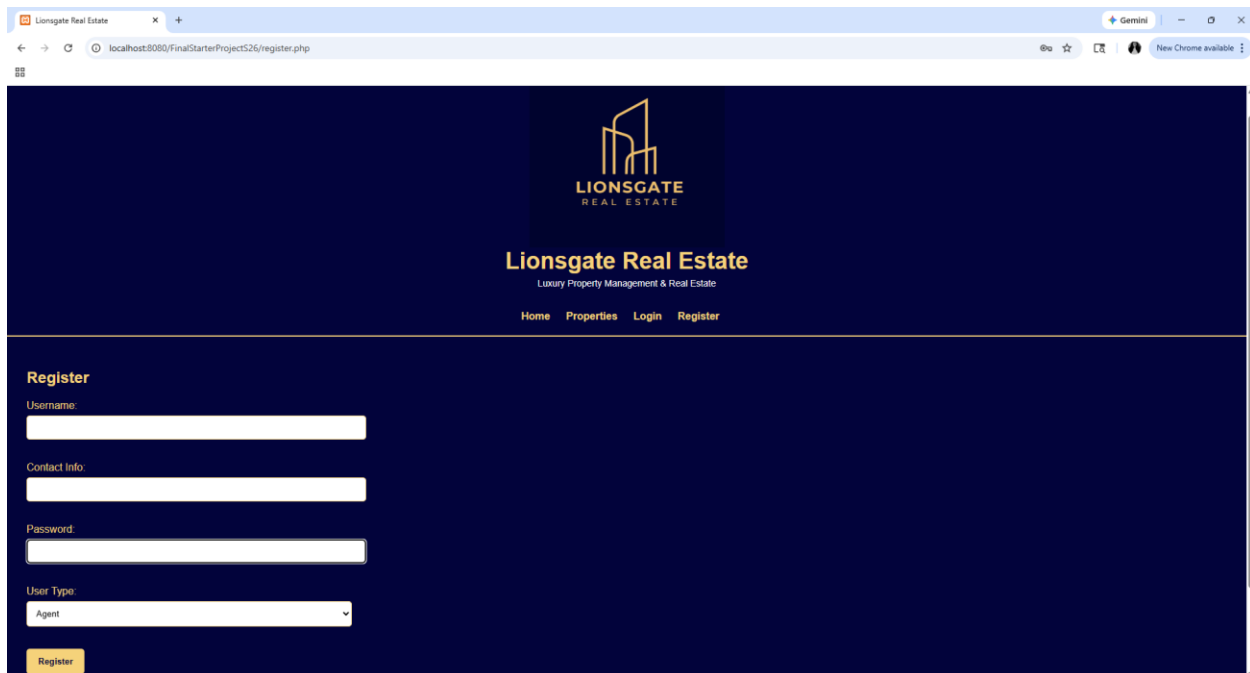
Solution

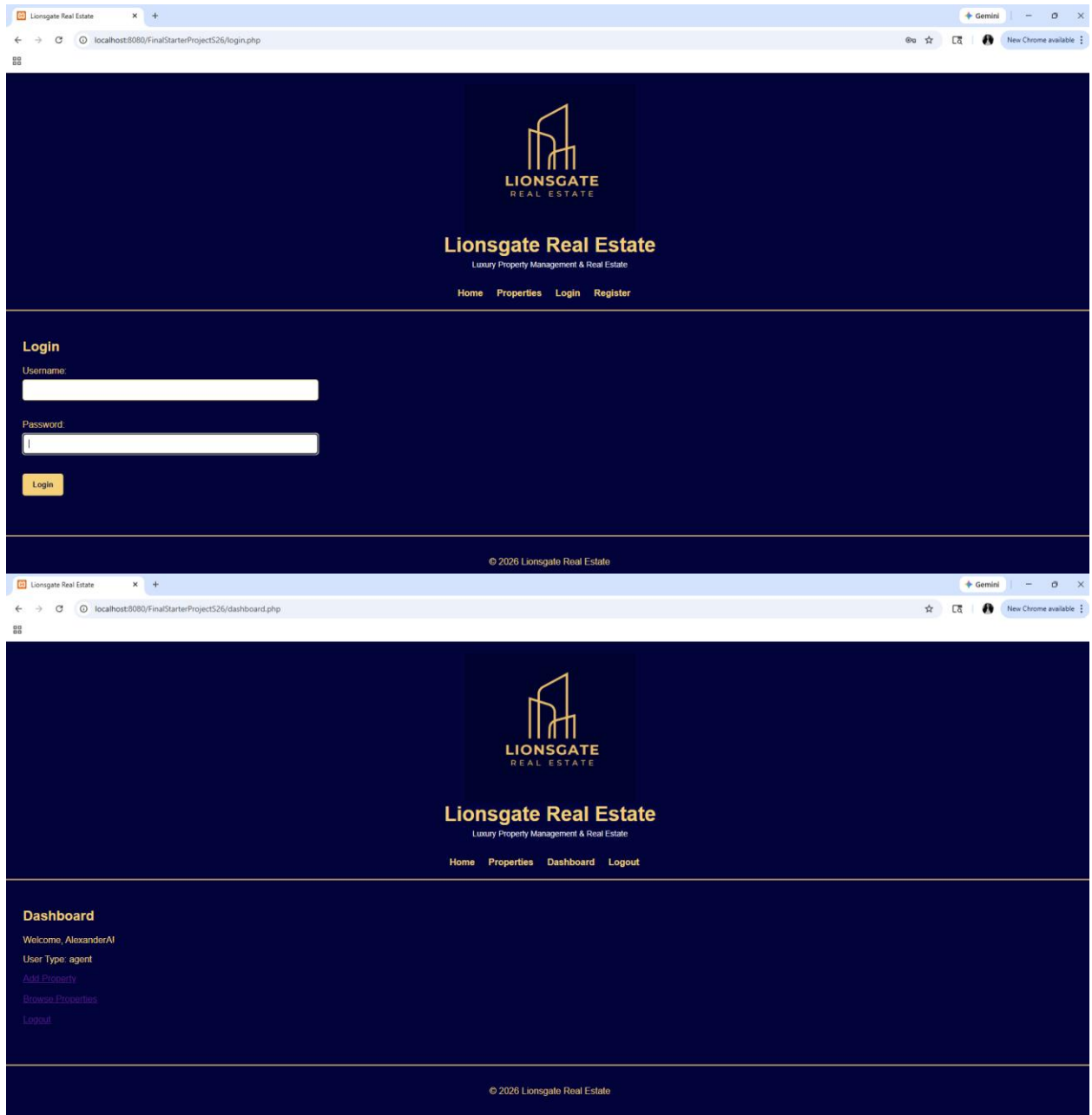
The website was redesigned using:

- Lionsgate Real Estate branding
- Custom logo made with Canva
- Dark navy luxury theme
- Gold color accents
- Updated moto for company added to home page
- Updated/Upgraded CSS styling

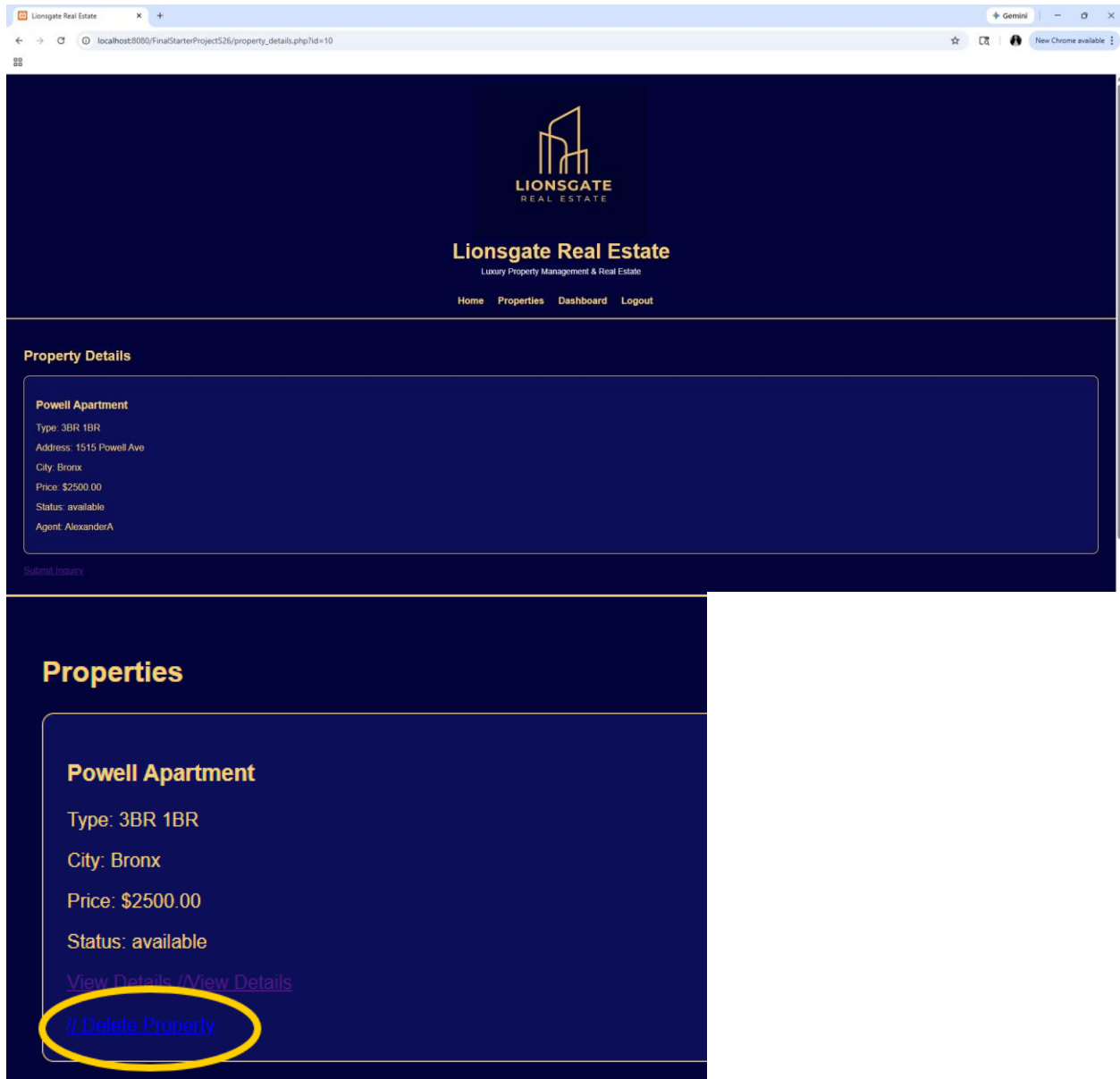
This created a modern luxury real estate appearance that would draw in consumers.

Screenshots of Platform









localhost8080 / 127.0.0.1 / real x + Gemini - O X

localhost8080/phpmyadmin/index.php?route=/database/structure&db=real_estate_portal_db

phpMyAdmin

Recent Favorites

- New
- information_schema
- mysql
- performance_schema
- phpmyadmin
- real_estate_portal_db
 - Procedures
 - New
 - AddOrUpdateUser
 - ProcessTransaction
 - Tables
 - New
 - favorites
 - inquiries
 - properties
 - Columns
 - Indexes
 - transactions
 - users
 - Views
 - New
 - propertylistingview
 - test

Server: 127.0.0.1:3307 Database: real_estate_portal_db

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers Tracking Designer Central columns

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
favorites	Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	64.0 K	0
inquiries	Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	64.0 K	0
properties	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	48.0 K	0
propertylistingview	Browse Structure Search Insert Edit View	~0	VIEW			
transactions	Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	64.0 K	0
users	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 K	0
6 tables	Sum	~19	InnoDB	utf8mb4_general_ci	272.0 K	0

Check all With selected:

Print Data dictionary

Create new table

Table name: Number of columns: 4 Create

localhost8080 / 127.0.0.1 / real x + Gemini - O X

localhost8080/phpmyadmin/index.php?route=/database/routines&type=PROCEDURE&db=real_estate_portal_db

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Server: 127.0.0.1:3307 Database: real_estate_portal_db

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers Tracking Designer Central columns

Routines

Check all Export Drop

Name Type Returns

- AddOrUpdateUser PROCEDURE Edit Execute Export Drop
- ProcessTransaction PROCEDURE Edit Execute Export Drop

localhost8080 / 127.0.0.1 / real x + Gemini - O X

localhost8080/phpmyadmin/index.php?route=/database/structure&tbl_type=table&db=real_estate_portal_db

phpMyAdmin

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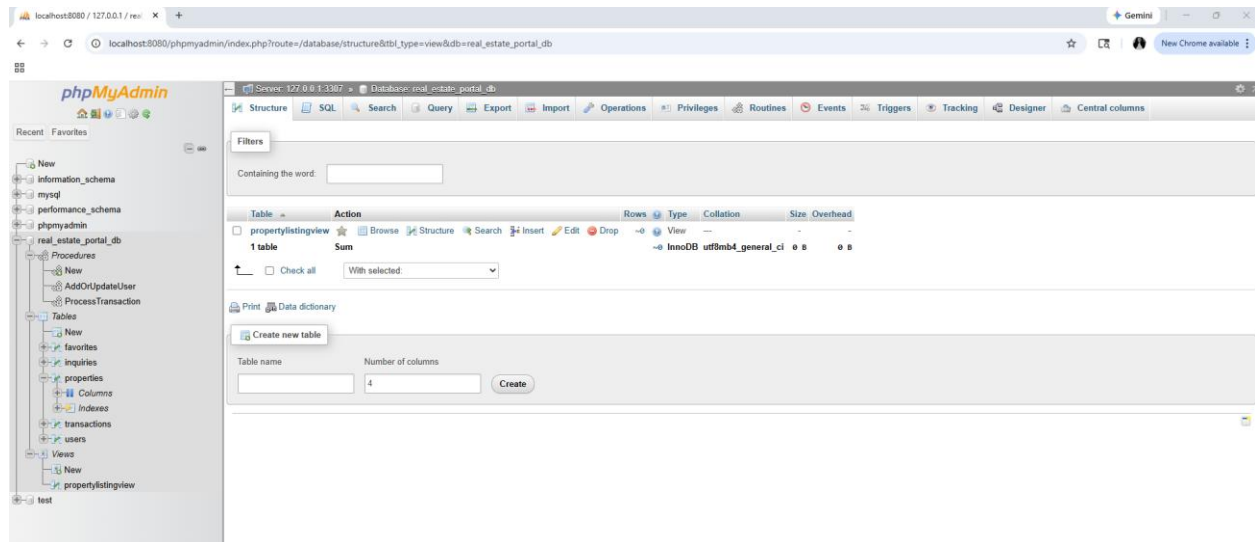
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5 tables	Sum	19	InnoDB	utf8mb4_general_ci	272.0 K	0

Check all With selected:

Print Data dictionary

Create new table

Table name: Number of columns: 4 Create



Conclusion

The Real Estate Agency Portal successfully demonstrates the development of a full-stack PHP and MySQL application with authentication, database management, CRUD functionality, role-based access control, and website customization.

The project evolved from a basic starter template into a branded luxury real estate platform featuring:

- Secure login
- Property management
- Inquiry System
- Delete property functionality
- Session handling
- Custom Lionsgate branding
- Luxury themed interface

The project provides a strong foundation for future enhancements such as:

- Image uploads
- Property search and filters
- Responsive mobile design
- Admin management
- Messaging systems
- Payment integration